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Abstract

Axial relative movement of an auxiliary member (30) and a hub body (22z) is performed, the hub body (22z) and an inner race (21) are combined with each other in an axial direction, and a part of the hub body (22z) deformed by a blade (33) of the auxiliary member (30) is disposed inside an engagement concave portion (26) of the inner race (21).

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 17/418,942, filed Jun. 28, 2021, which is a National Stage of International Application No. PCT/JP2020/004339 filed Feb. 5, 2020, claiming priority based on Japanese Patent Application No. 2019-018459, filed Feb. 5, 2019, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present invention relates to a staking assembly, a staking assembly manufacturing method, a hub unit bearing, a hub unit bearing manufacturing method, an automobile including a hub unit bearing, and an automobile manufacturing method.

BACKGROUND ART

(2) A vehicle wheel and a braking rotation body of an automobile are supported by a hub unit bearing to be rotatable with respect to a suspension device. FIG. 13 shows a structure of a hub unit bearing 1 described in U.S. Pat. No. 5,226,738. The hub unit bearing 100 has a configuration in which a hub 102 is rotatably supported on an inner radial side of an outer race 101 through a plurality of rolling elements 103. The outer race 101 includes a double row of outer race tracks 104a and 104b provided in an inner peripheral surface and a stationary flange 105 provided in an axially intermediate portion to support and fix the outer race 101 to a knuckle of the suspension device. The hub 102 includes a double row of inner race tracks 106a and 106b in an outer peripheral surface and includes a rotational flange 107 provided in an axial outer portion protruding outward in the axial direction in relation to the axial outer end surface of the outer race 101. The plurality of rolling elements 103 are arranged for each row between the double row of the outer race tracks 104a and 104b and the double row of the inner race tracks 106a and 106b in a rolling manner. With such a configuration, the hub 102 is rotatably supported on the inner radial side of the outer race 101.

(3) Additionally, the “outside” in the axial direction means the left side of FIG. 13 which is the outside of the vehicle body in a state in which the hub unit bearing 1 is assembled to the automobile. In contrast, the “inside” in the axial direction means the right side of FIG. 13 which is the center side of the vehicle body in a state in which the hub unit bearing 1 is assembled to the automobile.

(4) In the example shown in the drawings, the hub 102 is a combination of a hub body 108 and an inner race 109. The hub body 108 includes the inner race track 106a on the outside of the axial direction in the double row of the inner race tracks 106a and 106b in the outer peripheral surface of the axially intermediate portion and includes the rotational flange 107 in the axial outer portion. Further, the hub body 108 includes a fitting cylinder portion 110 having an outer diameter smaller than that of a portion adjacent to the outside of the axial direction in the axial inner portion existing on the inside of the axial direction in relation to the inner race track 106a on the outside of the axial direction.

(5) The inner race 109 includes the inner race track 106b on the inside of the axial direction in the double row of the inner race tracks 106a and 106b in the outer peripheral surface. Such an inner

race **109** is externally fitted to the fitting cylinder portion **110** while the axial outer end surface abuts against a step surface **111** existing in the axial outer end portion of the outer peripheral surface of the fitting cylinder portion **110**. In this state, the axial inner end surface of the inner race **109** is held down by a staking portion **112** formed by plastically deforming a cylindrical portion extending in the axial direction from the axial inner end portion of the fitting cylinder portion **110** outward in the radial direction. With such a configuration, the separation of the inner race **109** with respect to the hub body **108** is prevented.

(6) In the structure described in U.S. Pat. No. 5,226,738, if a force in which the staking portion **112** holds down the axial inner end surface of the inner race **109** is not sufficient, there is a possibility that relative slippage (displacement, creep) occurs between the hub body **108** and the inner race **109**.

(7) European Patent Application, Publication No. 0927651 describes a technique of preventing creep between the hub body and the inner race by roughening the surface roughness of a chamfered portion connecting the axial inner end surface and the inner peripheral surface of the inner race. Further, U.S. Pat. No. 5,822,860 describes a structure in which a chamfering dimension of a chamfered portion connecting the axial inner end surface and the inner peripheral surface of the inner race changes in the circumferential direction.

CITATION LIST

Patent Literature

(8) [Patent Literature 1]

(9) U.S. Pat. No. 5,226,738

[Patent Literature 2] European Patent Application, Publication No. 0927651

[Patent Literature 3] U.S. Pat. No. 5,822,860

SUMMARY OF INVENTION

Technical Problem

(10) However, even in the structure described in European Patent Application, Publication No. 0927651, if a force in which the staking portion holds down the axial inner end surface of the inner race is not sufficient, there is a possibility that creep occurs between the hub body and the inner race. Particularly, in the case of the structure in which a gap exists between the chamfered portion of the inner race and the staking portion, it is not possible to obtain the creep prevention effect due to the roughening of the surface roughness of the chamfered portion.

(11) In the structure described in U.S. Pat. No. 5,822,860, a force in which the staking portion holds down the axial inner end surface of the inner race becomes uneven in the circumferential direction. Accordingly, there is a possibility that the inner race may be distorted unevenly in the circumferential direction and the roundness of the inner race track may decrease. As a result, the bearing performance of the hub unit bearing may deteriorate.

(12) An object of the present invention is to realize a staking assembly capable of preventing displacement between a first member and a second member and a structure of a hub unit bearing capable of reliably preventing creep from occurring between a hub body and an inner race.

Solution to Problem

(13) According to an aspect of the present invention, there is provided a method of manufacturing a staking assembly including: preparing a first member, a second member including an engagement concave portion and a hole allowing the first member to be inserted therein, and an auxiliary member including a blade; combining the first member and the second member with each other in an axial direction; moving the auxiliary member and the first member relatively in an axial direction so that a part of the first member deformed by the blade of the auxiliary member in the axial relative movement is disposed inside the engagement concave portion of the second member; and forming a staking portion with respect to the second member in the first member.

(14) According to an aspect of the present invention, there is provided a method of manufacturing a staking assembly including: preparing a first member, a second member including a concave

portion and a hole allowing the first member to be inserted therein, and an auxiliary member; moving the auxiliary member and the first member relatively in an axial direction and including (a) combining the first member with the second member in the axial direction in accordance with the axial relative movement and (b) forming a convex portion engaging with the concave portion in accordance with the axial relative movement in the first member by the auxiliary member; and forming a staking portion with respect to the second member in the first member.

(15) According to an aspect of the present invention, there is provided a staking assembly including: a first member; and a second member including a hole allowing the first member to be inserted therein and combined with the first member, wherein the second member includes a concave portion, and wherein the first member includes a staking portion with respect to the second member, a convex portion provided in an outer surface of the first member and engaging with the concave portion of the second member, and a groove provided in the outer surface of the first member and extending from the convex portion toward a front end of the staking portion.

(16) According to an aspect of the present invention, there is provided a staking assembly including: a first member; and a second member including a hole allowing the first member to be inserted therein and combined with the first member, wherein the second member includes a concave portion, wherein the first member includes a staking portion with respect to the second member and a convex portion provided in an outer surface of the first member and engaging with the concave portion of the second member, wherein the concave portion of the second member includes a first surface intersecting an axial direction of the first member, wherein the convex portion of the first member includes a second surface parallel to the first surface of the concave portion, and wherein the first surface and the second surface are in close contact with each other in the axial direction.

(17) According to an aspect of the present invention, there is provided a hub unit bearing including: an outer race including an outer race track; a hub including an inner race track; and a plurality of rolling elements arranged between the outer race track and the inner race track, wherein the hub includes a hub body including an outer surface and an inner race disposed on the outer surface of the hub body and held by the hub body, wherein the inner race includes a concave portion, and wherein the hub body includes a staking portion with respect to the inner race, a convex portion provided in the outer surface of the hub body and engaging with the concave portion of the inner race, and a groove provided in the outer surface of the hub body and extending from the convex portion toward a front end of the staking portion.

(18) According to an aspect of the present invention, there is provided a method of manufacturing a hub unit bearing, wherein the hub unit bearing includes an outer race including an outer race track, a hub including an inner race track, and a plurality of rolling elements arranged between the outer race track and the inner race track and wherein the hub includes a hub body including an outer surface and an inner race disposed on the outer surface of the hub body and held by the hub body. The method includes: preparing the inner race including an engagement concave portion and an auxiliary member including a blade; combining the hub body and the inner race with each other in an axial direction; moving the auxiliary member and the hub body relatively in the axial direction so that a part of the hub body deformed by the blade in the axial relative movement is disposed inside the engagement concave portion of the inner race; and forming a staking portion with respect to the inner race in the hub body.

(19) According to an aspect of the present invention, there is provided a hub unit bearing including: an outer race including a double row of outer race tracks provided in an inner peripheral surface; a hub including a double row of inner race tracks provided in an outer peripheral surface; and a plurality of rolling elements which are arranged for each row between the double row of the outer race tracks and the double row of the inner race tracks in a rolling manner. The hub includes an inner race and a hub body. The inner race includes an inner race track on the inside of an axial direction in the double row of the inner race tracks on the outer peripheral surface. The hub body

includes an inner race track which is formed in an outer peripheral surface of an axially intermediate portion directly or through another member and is located on the outside of the axial direction in the double row of the inner race tracks, a fitting cylinder portion which exists on the inside of the axial direction in relation to the inner race track on the outside of the axial direction and to which the inner race is externally fitted, and a staking portion which is bent outward in a radial direction from an axial inner end portion of the fitting cylinder portion and holds down an axial inner end surface of the inner race. The inner race includes an engagement concave portion provided at one or a plurality of positions in a circumferential direction to open to the axial inner end surface and the inner peripheral surface. The hub body includes an engagement convex portion provided at one or a plurality of positions in the circumferential direction to engage with the engagement concave portion.

(20) The hub body can further include a concave groove which is recessed inward in the axial direction and extends outward in the radial direction from a portion which is close to the outside of the engagement convex portion in the radial direction and in which a phase in the circumferential direction coincides with that of the engagement convex portion in an axial outer surface of the staking portion.

(21) According to an aspect of the present invention, in order to manufacture the above-described hub unit bearing, there is provided a method of manufacturing the hub unit bearing including: press-fitting the inner race into the fitting cylinder portion of the hub body before forming the staking portion; allowing the engagement convex portion to engage with the engagement concave portion of the inner race while forming the engagement convex portion and the concave groove by pressing one or a plurality of positions in the circumferential direction of an outer peripheral surface of a radial outer portion of the cylindrical portion, extending inward in an axial direction from the fitting cylinder portion of the hub body before forming the staking portion, outward in the axial direction to be plastically deformed; and coupling and fixing the inner race and the hub body by plastically deforming the cylindrical portion outward in a radial direction to form the staking portion.

(22) It is preferable to perform the press-fitting and the engaging at the same time. In this case, the inner race can be press-fitted into the fitting cylinder portion of the hub body before forming the staking portion by pressing the inner race, externally fitted to an axial inner end portion of the hub body before forming the staking portion, outward in the axial direction by a portion deviating from a protrusion portion in a pressing punch including the protrusion portion protruding inward in the radial direction at one or a plurality of positions of the inner peripheral surface in the circumferential direction and the engagement convex portion can be allowed to engage with the engagement concave portion of the inner race while forming the engagement convex portion by pressing one or a plurality of positions in the circumferential direction of the radial outer portion of the cylindrical portion outward in the axial direction to be plastically deformed by the protrusion portion.

(23) Further, since the pressing punch includes a pressing surface on a front end surface, a gap can exist between a front end surface of the protrusion portion and a bottom surface of the engagement concave portion while the pressing surface comes into contact with a portion deviating from the engagement concave portion in an axial inner end surface of the inner race when performing the press-fitting and the engaging at the same time.

(24) It is preferable to further include confirming whether or not the engagement convex portion engages with the engagement concave portion after the engaging. It is possible to perform an operation of confirming whether or not the engagement convex portion engages with the engagement concave portion by, for example, with naked eyes or diagnosing images photographed by a camera.

(25) An automobile according to an aspect of the present invention includes the staking assembly or the hub unit bearing.

(26) In an automobile manufacturing method according to an aspect of the present invention, the hub unit bearing is manufactured by the hub unit bearing manufacturing method of the present invention.

Advantageous Effects of Invention

(27) According to the aspect of the present invention, for example, since the engagement convex portion of the hub body engages with the engagement concave portion of the inner race, it is possible to reliably prevent creep from occurring between the hub body and the inner race. Further, it is possible to prevent the staking assembly from being displaced between the first member and the second member.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a cross-sectional view showing a hub unit bearing according to an embodiment of the present invention.
- (2) FIG. 2 is a cross-sectional view showing a state seen from the left side of FIG. 1 when a braking rotation body and a stud are omitted.
- (3) FIG. 3 is a view corresponding to a cross-section X-X of FIG. 2, which shows a state before an axial outer end portion of an inner race is externally fitted to an axial inner end portion of a hub body before forming a staking portion and an axial inner end surface of the inner race is pressed by a pressing punch.
- (4) FIG. 4 is a view corresponding to a cross-section of X-X of FIG. 2, which shows a state in which the pressing punch is displaced downward from the state shown in FIG. 3 and a front end surface of the pressing punch comes into contact with the axial inner end surface of the inner race.
- (5) FIG. 5 is an enlarged view of a Y part of FIG. 4.
- (6) FIG. 6 is a view corresponding to a cross-section X-X of FIG. 2, which shows a state in which the pressing punch is further displaced downward from the state shown in FIG. 4 so that a press-fitting step and an engaging step are performed at the same time.
- (7) FIG. 7 is an enlarged view of a Z part of FIG. 6.
- (8) FIG. 8 is a view corresponding to a cross-section X-X of FIG. 2, which shows a state after the press-fitting step and the engaging step are completed.
- (9) FIG. 9 is a perspective view showing a state in which the press-fitting step and the engaging step are completed when viewed from the inside of the axial direction.
- (10) FIG. 10 is a view corresponding to a cross-section X-X of FIG. 2, which shows a state in which a hub is taken out.
- (11) FIG. 11(a) to FIG. 11(d) are schematic horizontal cross-sectional views showing an engagement concave portion of an inner race and an engagement convex portion of a hub body.
- (12) FIG. 12 is a partial schematic view of a vehicle including a hub unit bearing (bearing unit).
- (13) FIG. 13 is a cross-sectional view showing an example of a conventional structure of a hub unit bearing.

DESCRIPTION OF EMBODIMENTS

- (14) FIGS. 1 to 10 show an embodiment of the present invention. In the example of the drawings, a hub unit bearing 1 has a configuration in which a hub 3 is rotatably supported on the inner radial side of an outer race 2 through a plurality of rolling elements 4a and 4b. The hub unit 1 includes the outer race 2 which has outer race tracks 5a and 5b, the hub (staking assembly) 3 which has inner race tracks 11a and 11b, and a plurality of rolling elements 4a and 4b which are disposed between the outer race tracks 5a and 5b and the inner race tracks 7a and 7b.
- (15) The outer race 2 includes a double row of the outer race tracks 5a and 5b and a stationary flange 6. In an example, the outer race 2 is made of hard metal such as medium carbon steel. In

another example, the outer race **2** can be made of another material. The double row of the outer race tracks **5a** and **5b** are respectively formed in the inner peripheral surfaces of the axially intermediate portions of the outer race **2**. The outer race tracks **5a** and **5b** have a partially conical concave surface inclined in a direction in which the diameter increases toward a direction away from each other in the axial direction. The stationary flange **6** is formed in the axially intermediate portion of the outer race **2** to protrude outward in the radial direction. The stationary flange **6** includes a support hole **7** which is a screw hole provided at a plurality of positions of the radially intermediate portion in the circumferential direction. The outer race **2** is supported and fixed to a knuckle **8** in such a manner that a bolt **10** inserted through a passage hole **9** formed in the knuckle **8** constituting a suspension device is screwed into the support hole **7** of the stationary flange **6** from the inside of the axial direction and is further tightened.

(16) The hub (staking assembly, staking unit) **3** is disposed on the inner radial side of the outer race **2** to be coaxial with the outer race **2**. The hub **3** includes a double row of the inner race tracks **11a** and **11b** and a rotational flange **12**. The double row of the inner race tracks **11a** and **11b** are formed in a portion facing the double row of the outer race tracks **5a** and **5b** in the outer peripheral surface (outer surface) of the hub **3**. The double row of the inner race tracks **11a** and **11b** have a partially conical convex surface inclined in a direction in which the diameter increases toward a direction away from each other in the axial direction. The rotational flange **12** is formed in a portion located on the outside of the axial outer end portion of the outer race **2** in the axial direction of the hub **3** to protrude outward in the radial direction. The rotational flange **12** includes an attachment hole **13** which is provided at a plurality of positions of the radially intermediate portion in the circumferential direction to penetrate in the axial direction. In the example of the drawings, a serration portion formed near a base end of a stud **15** is press-fitted into the attachment hole **13** in order to couple and fix a braking rotation body **14** such as a disc or a drum to the rotational flange **12**. Further, an intermediate portion of the stud **15** is press-fitted into a passage hole **16** formed in the braking rotation body. Furthermore, a nut **19** is screwed into a male screw portion formed at a front end portion of the stud **15** and is further tightened while the male screw portion is inserted through a passage hole **18** formed in a wheel **17** in order to fix the wheel **17** constituting a vehicle wheel to the rotational flange **12**.

(17) Additionally, the “outside” in the axial direction means the left side of FIG. **1** and the lower side of FIGS. **3** to **10** corresponding to the outside of the vehicle body while the hub unit bearing **1** is assembled to the automobile. In contrast, the “inside” in the axial direction means the right side of FIG. **1** and the upper side of FIGS. **3** to **10** corresponding to the center side of the vehicle body while the hub unit bearing **1** is assembled to the automobile. Further, the attachment hole **13** is not shown in FIGS. **3**, **4**, **6**, **8**, and **10**.

(18) Each of the rolling elements **4a** and **4b** is disposed at a plurality of positions between the double row of the outer race tracks **5a** and **5b** and the double row of the inner race tracks **11a** and **11b** in a rolling manner while being held by cages **20a** and **20b**. In an example, each of the rolling elements **4a** and **4b** is made of hard metal such as bearing steel or ceramics. In another example, the rolling elements **4a** and **4b** can be made of another material. With such a configuration, the hub **3** is rotatably supported on the inner radial side of the outer race **2**. Further, in the example of FIG. **1**, tapered rollers whose outer diameter gradually increases from one end to the other end in the axial direction are used as the rolling elements **4a** and **4b**.

(19) The hub (staking assembly) **3** is substantially a combination of an inner race (second member) **21** and a hub body (first member) **22**. The hub **3** includes a hub body **22** having an outer peripheral surface (outer surface) **23a** and an inner race **21** disposed on the outer peripheral surface (outer surface) **23a** of the hub body **22** and held by the hub body **22**.

(20) The inner race **21** includes the inner race track **11b** on the inside of the axial direction on the outer peripheral surface in the double row of the inner race tracks **11a** and **11b**. In an example, the inner race **21** is made of hard metal such as bearing steel. In another example, the inner race **21** can

be made of another material.

(21) The hub body **22** includes the inner race track **11a** on the outside of the axial direction on the outer peripheral surface of the axially intermediate portion in the double row of the inner race tracks **11a** and **11b**. Further, the hub body **22** includes the rotational flange **12** which is provided in the outer portion in the axial direction and exists on the outside of the axial direction in relation to the inner race track **11a** on the outside of the axial direction. In an example, the hub body **22** is made of hard metal such as medium carbon steel. In another example, the hub body **22** can be made of another material. Further, the hub body **22** includes a fitting cylinder portion (cylinder portion) **23** which is provided in the axial inner portion existing on the inside of the axial direction in relation to the inner race track **11a** on the outside of the axial direction so that an outer diameter is smaller than that of a portion near the outside of the axial direction and the inner race **21** is externally fitted thereto. Furthermore, the hub body **22** includes a staking portion (swage portion, crimp portion) **24** which is bent outward in the radial direction from the axial inner end portion of the fitting cylinder portion **23** and holds down an axial inner end surface **34** of the inner race **21**. For example, the hub **3** is configured such that the inner race **21** is sandwiched from both side of the axial direction between the axial outer surface of the staking portion **24** and a step surface **25** facing the inside of the axial direction and existing in the axial outer end portion of the fitting cylinder portion **23** while the inner race **21** is externally fitted to the fitting cylinder portion **23** of the hub body **22** and the inner race **21** and the hub body **22** are coupled and fixed to each other. In an example, the hub body **22** includes the staking portion **24** (the staking portion **24** for holding the inner race **21**) with respect to the inner race **21**. The cylinder portion **23** of the hub body **22** includes a peripheral wall of which a thickness of at least a part changes along the axial direction. In the peripheral wall of the cylinder portion **23**, one end (first axial end) of the inner race **12** in the axial direction is disposed at a first portion (thick part) having a relatively large wall thickness and the other end (second axial end) of the inner race **21** in the axial direction is disposed at a second portion (thin part) having a relatively small wall thickness. The second portion of the peripheral wall of the hub body **22** is provided with the staking portion **24** having a bend extending in the circumferential direction and covering the axial end portion (second axial end) of the inner race **21**. The staking portion **24** includes a holding portion (staking flange (swage flange, crimp flange), ring edge) **24a** which extends in the radial direction and the circumferential direction and has a circular shape as a whole and a curved portion (bent portion, connection portion) **24b** which connects the radial inner end portion of the holding portion **24a** and the axial inner end portion of the cylinder portion **23** (FIG. 10).

(22) In this embodiment, a plurality of concave portions (engagement concave portions, concave grooves, recesses, engagement recesses, key grooves) **26** provided in the inner race **21** respectively engage with a plurality of convex portions (engagement convex portions, protrusions, keys) **27** provided in the hub body **22** so that the relative rotation (creep) between the inner race **21** and the hub body **22** is prevented. The engagement concave portions **26** are formed at a plurality of positions (in the example shown in the drawings, four positions) in the circumferential direction of the inner race **21** so as to open to the inner peripheral surface of the axial inner end portion and the axial inner end surface **34** of the inner race **21**. In other words, the engagement concave portion **26** opens inward in the radial direction and outward in the axial direction. The engagement convex portion **27** is formed at a plurality of positions in the circumferential direction of the axial inner end portion of the fitting cylinder portion **23** of the hub body **22** to protrude outward in the radial direction. For example, as will be described later, the engagement convex portion **27** can be formed by subjecting an outer peripheral surface of a cylindrical portion **28** constituting a hub body **22z** before forming the staking portion **24** to a protruding process (local shaving process). As shown in FIG. 10, the staking portion **24** is formed by plastically deforming the cylindrical portion **28** outward in the radial direction. The phase in the circumferential direction of the axial outer surface of the staking portion **24** coincides with that of the engagement convex portion **27**. The phase in the

circumferential direction of the concave portion **26** of the inner race **21** coincides with that of the engagement convex portion **27**. A concave groove **29** which extends outward in the radial direction and is recessed inward in the axial direction is formed over the radial direction from a portion adjacent to the outside of the engagement convex portion **27** in the radial direction.

(23) In other words, the hub body **22** has a shaft shape along a predetermined axial direction. The inner race **21** includes a hole **120** into which the hub body **22** is inserted and the concave portion **26** which is provided in a part of the circumferential direction in the wall surface (inner surface, inner peripheral surface) facing the hole **120**. The hub body **22** includes the convex portion **27** which is provided in a part of the outer surface **23a** in the circumferential direction and engages with the concave portion **26** of the inner race **21** and the groove (processed groove, processing mark) **29** which is provided in the outer surface **23a** and extends from the convex portion **27** toward the front end of the staking portion **24**. One end (first end) of the groove **29** in the extension direction is located at one end of the convex portion **27** or in the vicinity thereof in the hub body **22**. The other end (second end) of the groove **29** in the extension direction is disposed at a position closer to the front end of the staking portion **24** than the convex portion **27**. In an example, the extension direction of the groove **29** is parallel to the center axis of the hub body **22**. The convex portion **27** and the groove **29** are linearly arranged side by side on one line along the extension direction of the groove **29**. In an example, the first end and the second end of the groove **29** may have different step shapes in the extension direction. The groove **29** is provided in both the holding portion **24a** and the curved portion **24b** of the staking portion **24**. In another example, the groove **29** is provided in the curved portion **24b** of the staking portion **24**. In another example, the groove **29** is not provided in the staking portion **24**. The groove **29** provided in the staking portion **24** can be advantageous in preventing the staking portion **24** from being broken or cracked in forging (plastic deforming, heading, press forming, bending) for forming the staking portion **24**.

(24) The concave portion **26** of the inner race **21** includes a bottom surface (first surface) **126** which intersects (for example, perpendicular to) the axial direction of the hub body **22** (FIG. 7). The convex portion **27** of the hub body **22** includes a facing surface (second surface) **127** which is parallel to the first surface **126** of the concave portion **26**. The first surface **126** and the second surface **127** are in close contact with each other in the axial direction. The number of the concave portions **26** arranged in the circumferential direction can be, for example, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more. The number of the convex portions **27** arranged in the circumferential direction can be, for example, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more. The inner race **21** includes a plurality of recesses which are arranged as the concave portions **26** to be away from each other. For example, the plurality of recesses include a first recess **26A** and a second recess **26B** which are adjacent to each other (FIG. 9). In an example, the distance between the first recess **26A** and the second recess **26B** (the peripheral surface length between two adjacent recesses in the circumferential direction (hereinafter, peripheral surface length)) is twice or more the horizontal width (the width (hereinafter, recess width) of the recess in the circumferential direction) of the first recess **26A** (or the second recess **26B**). For example, the recess width can be set to $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$, $\frac{1}{6}$, $\frac{1}{7}$, $\frac{1}{8}$, $\frac{1}{9}$, $\frac{1}{10}$, or less of the peripheral surface length. An area provided with the concave portion **26** on the inner surface of the inner race **26** is a partial area. For example, the circumferential length of the area provided with the concave portion **26** on the inner surface of the inner race **26** can be 30%, 25%, 20%, 15%, 10%, or 5% or less of the entire circumferential length.

(25) The hub (staking assembly) **3** and the hub unit bearing **1** described above can be manufactured as below. First, the inner race **21** including the inner race track **11b** on the inside of the axial direction and the plurality of engagement concave portions **26** is obtained by subjecting a metal material to necessary processing such as forging or grinding. For example, the inner race **21** can be obtained by forging a metal material to form the rough shape of the inner race **21** and then performing a finishing process such as grinding on the outer peripheral surface to form the inner race track **11b** on the inside of the axial direction. The engagement concave portion **26** can be

formed at the same time when the rough shape of the inner race **21** is formed by forging the metal material. Alternatively, the engagement concave portion **26** can be formed by cutting after forming the rough shape of the inner race **21**. As will be described later, the convex portion **27** engaging with the concave portion **26** is disposed in the concave portion **26** along with the formation of the convex portion **27**. In this procedure, for example, the volume of the convex portion **27** disposed in the concave portion **26** gradually increases. This engagement technique has advantages of eliminating the need for high dimensional accuracy for the concave portion **26**. For example, in the formation of the concave portion **26**, cutting can be avoided and forging with dies can be easily applied.

(26) Further, the hub body **22z** before forming the staking portion **24** shown in FIG. **3** is obtained by performing necessary processing such as forging or grinding on a metal material. That is, the hub body **22z** includes the inner race track **11a** which is on the outside of the axial direction, the rotational flange **12**, the fitting cylinder portion **23** and the step surface **25**, and the cylindrical portion **28** extending inward in the axial direction from the axial inner end portion of the fitting cylinder portion **23**. For example, the outer peripheral surface of the fitting cylinder portion **23** and the outer peripheral surface of the cylindrical portion **28** exist on the same cylindrical surface. The hub body **22z** can be obtained by forging a metal material to form the rough shape of the hub body **22z** and performing a finishing process such as grinding on the outer peripheral surface.

(27) Next, the hub body **22z** is supported on a support base (not shown) with the axial outer end portion facing downward and the axial inner end portion facing upward. Then, the rolling element **4a** of the outer row in the axial direction is disposed around the inner race track **11a** on the outside of the axial direction in the hub body **22z** to be held by the cage **20a** on the outside of the axial direction. Further, the outer race **2** is disposed around the axially intermediate portion of the hub body **22z**. Additionally, the outer race **2**, the rolling elements **4a** and **4b**, and the cages **20a** and **20b** are omitted in FIGS. **3** to **10**.

(28) Next, the rolling element **4b** of the inner row in the axial direction is disposed around the inner race track **11b** on the inside of the axial direction in the inner race **21** to be held by the cage **20b** on the inside of the axial direction. Then, as shown in FIG. **3**, the axial outer end portion of the inner race **21** is externally fitted (lightly press-fitted) to the axial inner end portion of the cylindrical portion **28** of the hub body **22z**. At this time, the engagement concave portion **26** of the inner race **21** and a protrusion portion (blade) **33** of the pressing punch **30** disposed above the support base to be displaced in the vertical direction are set such that the phases in the circumferential direction coincide with each other.

(29) Next, as shown in FIGS. **3** and **4**, the pressing punch **30** is lowered so that a pressing surface **31** provided in the pressing punch **30** comes into contact with the axial inner end surface (axial end surface) **34** of the inner race **21**. The protrusion portion (blade) **33** of the pressing punch **30** is aligned to the engagement concave portion **26** of the inner race **21** in the circumferential direction. The pressing punch **30** includes a concave portion **32** which opens to the center portion of the lower surface (front end surface) and has a circular shape when viewed from below and includes the pressing surface **31** which is provided around the concave portion **32** in the lower surface. Further, the pressing punch **30** includes the protrusion portion **33** which is provided at a plurality of positions (for example, four positions) in the circumferential direction of the inner peripheral surface of the concave portion **32** to protrude inward in the radial direction. In an example, the protrusion portion **33** is formed in the inner peripheral surface of the concave portion **32** over the axial direction and the lower end portion (front end portion) protrudes downward in relation to the pressing surface **31**. Further, the protrusion portion **33** includes a blade on the inner peripheral edge of the lower end surface and both side edges in the circumferential direction for performing a protruding process (local shaving process) on the outer peripheral surface of the cylindrical portion **28** as described later.

(30) As shown in FIG. **5**, the lower end portion of the protrusion portion **33** is disposed inside the

engagement concave portion **26** while the pressing surface **31** of the pressing punch **30** comes into contact with the axial inner end surface **34** of the inner race **21**. Further, the lower end surface of the protrusion portion **33** faces the bottom surface (the surface facing the inside of the axial direction) of the engagement concave portion **26** with a gap therebetween. In other words, when the phases in the circumferential direction (the circumferential positions) between the protrusion portion **33** of the pressing punch **30** and the engagement concave portion **26** of the inner race **21** coincide with each other, the pressing surface **31** of the pressing punch **30** comes into contact with the axial inner end surface **34** of the inner race **21**. In contrast, when the phases in the circumferential direction between the protrusion portion **33** of the pressing punch **30** and the engagement concave portion **26** of the inner race **21** are displaced, the lower end surface of the protrusion portion **33** contacts the axial inner end surface **34** of the inner race **21** and a gap is formed between the pressing surface **31** and the axial inner end surface **34** of the inner race **21**. Thus, it is possible to determine whether or not the phases in the circumferential direction between the protrusion portion **33** of the pressing punch **30** and the engagement concave portion **26** of the inner race **21** coincide with each other by determining whether or not the pressing surface **31** of the pressing punch **30** is in contact with the axial inner end surface **34** of the inner race **21** with naked eyes or by diagnosing images photographed by a camera. Alternatively, specifically, for example, as will be described later, it may be determined whether or not the phases in the circumferential direction between the protrusion portion **33** of the pressing punch **30** and the engagement concave portion **26** of the inner race **21** coincide with each other depending on the load applied to the pressing punch **30** when press-fitting the inner race **21** into the fitting cylinder portion **23** of the hub body **22z** and/or forming the engagement convex portion **27** by a protruding process (local shaving process). In another example, the phases (circumferential positions) of the concave portion **26** and the protrusion portion **33** may be aligned by relatively rotating one of the pressing punch **30** and the inner race **21** (the hub body **22z**) during press-fitting.

(31) Additionally, the pressing punch **30** including the protrusion portion **33** can be integrally formed as a whole and the protrusion portion **33** can be provided by supporting and fixing a separate member to the inner peripheral surface of the concave portion **32** provided in the body portion. If the protrusion portion **33** is provided by supporting and fixing the separate member to the body portion, it is possible to easily separate the separate member and polish or replace the separate member when the blade provided in both side edges in the circumferential direction and the inner peripheral edge of the lower end surface of the protrusion portion **33** is worn out. When the blade is a right-angled blade, surface grinding can be easily applied to polishing. By polishing, the corners of the blade can be edged. Alternatively, the blade of the protrusion portion **33** can be replaced with another blade having an edge if necessary. At least one of the shape and the material of the protrusion portion (blade) **33** is set in response to the physical characteristics (material and heat treatment) of the hub body **22z**. For example, an optimum blade can be attached to the protrusion portion **33** in response to the physical characteristics of the hub body **22z**.

(32) Further, when a positioning convex portion protruding downward is formed in the lower surface of the pressing punch **30** separately from the protrusion portion **33**, a positioning concave portion protruding outward in the axial direction is formed in the axial inner end surface **34** of the inner race **21** separately from the engagement concave portion **26**, and the positioning convex portion and the positioning concave portion engage with each other while the pressing surface **31** of the pressing punch **30** is in contact with the axial inner end surface **34** of the inner race **21**, the phases in the circumferential direction between the inner race **21** and the pressing punch **30** can be allowed to coincide with each other. In this case, it is preferable to regulate the dimension of each portion so that a gap exists between the lower end surface of the protrusion portion **33** and the bottom surface of the engagement concave portion **26** when the front end surface (the lower end surface) of the positioning convex portion comes into contact with the bottom surface (the surface facing the inside of the axial direction) of the positioning concave portion while the pressing

surface **31** of the pressing punch **30** is in contact with the axial inner end surface **34** of the inner race **21**.

(33) Next, as shown in FIGS. **4** to **6**, a press-fitting step and an engaging step are performed at the same time. In the press-fitting step, the pressing punch **30** is further lowered so that the inner race **21** is press-fitted into the fitting cylinder portion **23** of the hub body **22z**. The inner race **21** and the hub body **22z** are combined (assembled, fitted) in the axial direction. In the engaging step, the engagement convex portion **27** is allowed to engage with the engagement concave portion **26** while forming the engagement convex portion **27** by forming a concave groove **29z**. In the initial stage of the axial relative movement between the pressing punch **30** and the hub body **22z**, (a) the pressing surface **31** of the pressing punch **30** comes into contact with the inner race **21**, (b) at least a part of the protrusion portion (blade) **33** of the pressing punch **30** is disposed inside the engagement concave portion **26** of the inner race **21**, and (c) an axial gap is provided between the protrusion portion (blade) **33** and the inner surface of the engagement concave portion **26** (FIG. **5**). In accordance with the axial relative movement between the pressing punch **30** and the hub body **22z**, the axial relative movement of the pressing punch (auxiliary member) **30** and the hub body **22z** is performed. That is, the axial inner end surface **34** of the inner race **21** is pressed outward (downward) in the axial direction by the pressing surface **31** until the outer end surface of the inner race **21** in the axial direction collides with the step surface **25**. Accordingly, the inner race **21** is press-fitted into the fitting cylinder portion **23** of the hub body **22z**. At the same time, a plurality of positions in the circumferential direction of the radial outer portion of the cylindrical portion **28** of the hub body **22z** are pressed outward in the axial direction by the lower end surface of the protrusion portion **33** to be plastically deformed (so that a protruding process (local shaving process) is performed). As shown in FIG. **7**, an extra thickness generated as a result of forming the concave groove **29z** is crushed between the lower end surface of the protrusion portion **33** and the bottom surface of the engagement concave portion **26** and the engagement convex portion **27** protruding outward in the radial direction is formed at a plurality of positions in the circumferential direction of the axial inner end portion of the fitting cylinder portion **23**. Further, the engagement convex portion **27** is disposed inside the engagement concave portion **26** of the inner race **21** and the engagement convex portion **27** engages with the engagement concave portion **26**. That is, a part of the hub body **22z** deformed (shaved, partially shaved) by the protrusion portion (blade) **33** of the pressing punch **30** in the axial relative movement between the pressing punch **30** and the hub body **22z** is disposed inside the engagement concave portion **26** of the inner race **21**. Accordingly, the creep (displacement) between the inner race **21** and the hub body **22z** is prevented. In an example, as shown in FIG. **7**, a gap exists between the pressing surface **31** and the axial inner end surface **34** of the inner race **21** in a completed state of the press-fitting step and the engaging step in which the pressing punch **30** is lowered to a bottom dead center. However, the pressing surface **31** and the axial inner end surface **34** of the inner race **21** can be brought into contact with each other in the completed state of the press-fitting step and the engaging step. Additionally, a groove (processed groove, processing mark) **29z** opening to the axial inner end surface is formed in a portion in which the phase in the circumferential direction in the outer peripheral surface of the cylindrical portion **28** coincides with that of the engagement convex portion **27** as the engagement convex portion **27** is formed by a protruding process (local shaving process). In addition, the inner race **21** can be press-fitted into the hub body **22z** with a relatively light force (light press-fitting). This is because the displacement between the inner race **21** and the hub body **22** is suppressed by the engagement convex portion **27**. For example, the press-fitting margin can be set to a value close to zero.

(34) Next, as shown in FIGS. **6** to **8**, the pressing punch **30** is raised (retracted). Then, it is confirmed whether or not the engagement concave portion **26** of the inner race **21** engages with the engagement convex portion **27** of the hub body **22z** with naked eyes or diagnosing images photographed by a camera.

(35) When the engagement concave portion **26** and the engagement convex portion **27** engage with

each other, as shown in FIG. 10, the staking portion 24 is formed by plastically deforming the cylindrical portion 28 of the hub body 22z outward in the radial direction. In this way, the hub unit bearing 1 of this example is obtained by processing the hub body 22z into the hub body 22 and coupling and fixing the inner race 21 and the hub body 22 to each other.

(36) Additionally, various methods known from the past can be used for the method of processing the cylindrical portion 28 into the staking portion 24. For example, a die having a center axis inclined with respect to the center axis of the hub body 22z supported by the support base is pressed against the cylindrical portion 28. In this state, the staking portion 24 can be formed by swaging in which the die is swung around the center axis of the hub body 22z like a locus of the center axis due to precession. Alternatively, the staking portion 24 may be formed by flat-pressing in which a mold is pressed in the axial direction.

(37) Further, the process of manufacturing the hub unit bearing 1 can be changed in order as appropriate or changed to be carried out simultaneously or separately as long as there is no contradiction. That is, for example, when the inner race 21 is press-fitted into the fitting cylinder portion 23 of the hub body 22z and a plurality of positions in the circumferential direction of the radial outer portion of the cylindrical portion 28 of the hub body 22z is subjected to a protruding process (local shaving process), the engagement convex portion 27 can engage with the engagement concave portion 26 of the inner race 21 while forming the engagement convex portion 27.

(38) In the hub unit bearing 1 of this embodiment, since the engagement concave portion 26 of the inner race 21 engages with the engagement convex portion 27 of the hub body 22, it is possible to reliably prevent creep from occurring between the inner race 21 and the hub body 22.

(39) Further, in this embodiment, it is possible to satisfactorily ensure the bearing performance of the hub unit bearing 1. That is, when the axial inner end surface of the inner race is strongly pressed by the staking portion in order to prevent the creep between the hub body and the inner race, the inner race is elastically deformed so that the inner race track on the inside of the axial direction expands. In contrast, in this embodiment, since the creep between the inner race 21 and the hub body 22 is prevented due to the engagement between the engagement concave portion 26 of the inner race 21 and the engagement convex portion 27 of the hub body 22, there is no need to excessively increase a force in which the axial inner end surface 34 of the inner race 21 is held down by the staking portion 24 to prevent the occurrence of creep. That is, the staking portion 24 may suffice to hold the axial inner end surface 34 of the inner race 21 with a force sufficient to prevent the inner race 21 from being displaced inward in the axial direction. Therefore, the elastic deformation amount of the inner race 21 can be suppressed to be small and the bearing performance can be satisfactorily ensured. Further, since there is no need to excessively increase a force in which the staking portion 24 holds down the axial inner end surface 34 of the inner race 21 and the processing load when forming the staking portion 24 can be suppressed to be small, a processing machine can be decreased in size.

(40) Further, in this example, the engagement convex portion 27 engages with the engagement concave portion 26 opening to the axial inner end surface 34 of the inner race 21 and the inner peripheral surface of the axial inner end portion. Thus, it is possible to easily confirm whether or not the engagement concave portion 26 engages with the engagement convex portion 27 in the state before forming the staking portion 24, that is, whether or not creep between the inner race 21 and the hub body 22z is reliably prevented with naked eyes or diagnosing images photographed by a camera. Therefore, it is possible to suppress the occurrence of defective products and improve the yield.

(41) Additionally, in this example, each of the engagement concave portion 26 and the engagement convex portion 27 is formed at four positions, but the number of the engagement concave portions and the engagement convex portions is not particularly limited and can be one to three or five or more as long as the creep between the inner race and the hub body can be prevented. Further, the

shapes and dimensions of the engagement concave portion **26** and the engagement convex portion **27** are not particularly limited as long as the creep between the inner race and the hub body can be prevented, the engagement convex portion can be formed by a protruding process (local shaving process), and the strength and rigidity of the inner race and the hub body can be sufficiently ensured.

(42) The rolling elements **4a** and **4b** are not limited to tapered rollers. Balls can also be used as the rolling elements **4a** and **4b**. Further, in the present invention, the hub body **9a** is not limited to a hub unit bearing for a solid driven wheel, but can be also applied to a hub unit bearing for a drive wheel having an engagement hole such as a spline hole for engaging a drive shaft enabling torque transmission in the center of the hub body.

(43) FIG. **11(a)** to FIG. **11(d)** are schematic horizontal cross-sectional views showing the engagement concave portion **26** of the inner race **21** and the engagement convex portion **27** of the hub body **22**. In the example of FIG. **11(a)**, the convex portion **27** contacts the wall surface of the concave portion **26** in the radial direction and the circumferential direction. In the example of FIG. **11(b)**, the convex portion **27** contacts the wall surface of the concave portion **26** in the circumferential direction. There is a gap between the wall surface of the concave portion **26** and the convex portion **27** in the radial direction. In the example of FIG. **11(c)**, the convex portion **27** contacts the wall surface of the concave portion **26** in the radial direction. There is a gap between the wall surface of the concave portion **26** and the convex portion **27** in the circumferential direction. In the example of FIG. **11(d)**, the convex portion **27** is formed such that a gap exists between the wall surface of the concave portion **26** and the convex portion **27** in the radial direction and the circumferential direction. When the contact area of the wall surface of the concave portion **26** and the convex portion is large, the displacement between the inner race **21** and the hub body **22** is reliably prevented. When the contact area of the wall surface of the concave portion **26** and the convex portion is small, an excessive stress is prevented. Additionally, the convex portion **27** formed by a protruding process (local shaving process) can have a higher hardness than other portions of the hub body **22**.

(44) FIG. **12** is a partial schematic view of a vehicle **200** including a hub unit bearing (bearing unit) **151**. The present invention can be applied to any one of a hub unit bearing for a drive wheel and a hub unit bearing for a driven wheel. In FIG. **12**, a hub unit bearing **151** is for a drive wheel and includes an outer race **152**, a hub **153**, and a plurality of rolling elements **156**. The outer race **152** is fixed to a knuckle **201** of a suspension device by a bolt and the like. A vehicle wheel (and a braking rotation body **22**) **202** is fixed to a flange (rotational flange) **153A** provided in the hub **153** using a bolt and the like. Further, the vehicle **200** can have the above-described support structure for the hub unit bearing **151** for a driven wheel.

(45) The present invention is not limited to the hub of the hub unit bearing and can be also applied to other staking assemblies (staking units) in which a first member and a second member having a hole into which the first member is inserted are combined.

REFERENCE SIGNS LIST

(46) **1** Hub unit bearing **2** Outer race **3** Hub (staking assembly, staking unit) **4a, 4b** Rolling element **5a, 5b** Outer race track **6** Stationary flange **7** Support hole **8** Knuckle **9** Passage hole **10** Bolt **11a, 11b** Inner race track **12** Rotational flange **13** Attachment hole **14** Braking rotation body **15** Stud **16** Passage hole **17** Wheel **18** Passage hole **19** Nut **20a, 20b** Cage **21** Inner race (second member) **22, 22z** Hub body (first member) **23** Fitting cylinder portion **24** Staking portion **25** Step surface **26** Engagement concave portion (concave portion) **27** Engagement convex portion **28** Cylindrical portion **29, 29z** Concave groove (groove, processed groove, processing mark) **30** Pressing punch (auxiliary member) **31** Pressing surface **32** Concave portion **33** Protrusion portion (blade) **34** Axial inner end surface **100** Hub unit bearing **101** Outer race **102** Hub **103** Rolling element **104a, 104b** Outer race track **105** Stationary flange **106a, 106b** Inner race track **107** Rotational flange **108** Hub

body **109** Inner race **110** Fitting cylinder portion **111** Step surface **112** Staking portion **120** Hole
126 Bottom surface (first surface) **127** Facing surface (second surface)

Claims

1. A staking assembly comprising: a first member; and a second member including a hole and combined with the first member so that the first member is inserted into the hole of the second member, wherein the second member includes a concave portion, and wherein the first member includes a staking portion, a convex portion engaging with the concave portion of the second member, and a groove extending at least in a radial direction from the convex portion toward an end of the staking portion.
 2. The staking assembly according to claim 1, wherein the second member includes a plurality of recesses each forming a concave portion, wherein the plurality of recesses include a first recess and a second recess, and wherein a distance between the first recess and the second recess is twice or more a horizontal width of the first recess or the second recess.
 3. An automobile comprising: a staking assembly according to claim 1; and a vehicle body to which the staking assembly is assembled.
 4. A hub unit bearing comprising: an outer race including an outer race track; a hub including an inner race track; and a plurality of rolling elements arranged between the outer race track and the inner race track, wherein the hub includes a hub body including an outer surface and an inner race disposed on the outer surface of the hub body and held by the hub body, wherein the inner race includes a concave portion, and wherein the hub body includes a staking portion, a convex portion engaging with the concave portion of the inner race, and a groove extending at least in a radial direction from the convex portion toward an end of the staking portion.
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